

Specific Factors and Income Distribution

ECON 308 International Economic Relations | Chapter 4

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Opening Puzzle: Why Do Steelworkers Oppose Free Trade?

- In 2018, the US imposed a **25% tariff on Canadian steel**. Hamilton steelworkers cheered. Canadian auto workers groaned.
- The Ricardian model says *both* countries gain from trade. So why such fierce political opposition?
- The Ricardian model is **too simple**: it has one factor (labour) and no distributional conflict.
- This chapter adds **capital** and **land** — and suddenly trade creates *real* winners and losers.

The Distributional Question

Trade raises national income but the gains are not shared equally. The Specific Factors Model explains *who* gains and *who* loses — and why they lobby so differently.

From Ricardo to Reality

Ricardian model: one factor.

Everyone gains. No distributional conflict.

Specific Factors model: three factors.

Capital and land are *sector-specific* in the short run. Trade **hurts** owners of capital in the import-competing sector.

This is why the political economy of trade is contentious even when overall welfare rises.

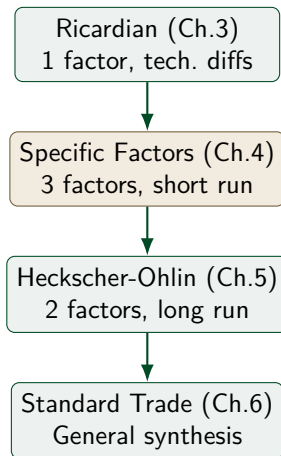
Where We Are in the Course

- **Ch. 2** (Lecture 1): Trade facts and the gravity model.
- **Ch. 3** (Lecture 2): *Why* trade? Comparative advantage; all gain (Ricardian).
- **Ch. 4** (today): *Who gains and who loses?* Short-run specificity of capital and land.
- **Ch. 5** (next): Long-run — both factors mobile. H-O model; factor price equalisation.

The two-step logic:

The Specific Factors model is the *short-run* version of the Heckscher-Ohlin model. As capital and land become mobile (long run), SF converges to H-O.

Ch. 4 = short run; Ch. 5 = long run.



Learning Objectives

After this lecture you will be able to:

1. State the assumptions of the specific factors model; explain what “specific” means and why it matters.
2. Derive MPL_M and MPL_F and write the labour market equilibrium condition.
3. Solve analytically for the equilibrium wage and labour allocation given production functions, prices, and endowments.
4. Predict how a rise in P_M affects wages, capital returns, and land returns.
5. Explain why labour’s real wage effect is *ambiguous* while specific-factor effects are *unambiguous*.
6. Apply the model to **Canadian examples**: Alberta oil boom, Ontario manufacturing, dairy supply management, immigration.
7. Connect the specific-factors model to the political economy of trade protection.

Krugman, Obstfeld & Melitz, *International Economics*, 12th ed., Chapter 4.

Model Setup: Three Factors, Two Sectors

Two goods:

- **Manufactures (M)**: uses capital K and labour L_M .
- **Food (F)**: uses land T and labour L_F .

Three factors:

- **Labour L** : **mobile** — moves freely between sectors until wages equalise.
- **Capital K** : **specific** to M only.
- **Land T** : **specific** to F only.

Why “specific”?

A steel mill cannot become a wheat farm overnight. In the short run, capital and land are locked into their sector. This is the key departure from the Ricardian model.

Variable list:

$L = L_M + L_F$ = total labour endowment

K = capital (specific to M)

T = land (specific to F)

Q_M, Q_F = outputs

P_M, P_F = goods prices

w = wage (equalises across sectors)

$r_K = P_M \cdot MPK_M$ = return to capital

$r_T = P_F \cdot MPT_F$ = return to land

The Ricardian Bridge

Ricardian model: 1 factor (labour), linear PPF, no distributional conflict.

Specific Factors: 3 factors, concave PPF, *some* factors lose from trade.

Production Functions and Marginal Products

Production functions:

$$Q_M = F(K, L_M), \quad Q_F = G(T, L_F)$$

Both exhibit:

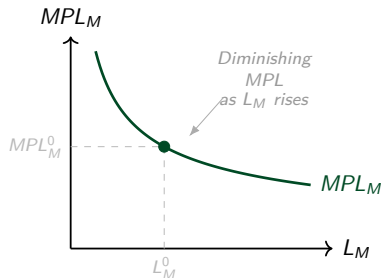
- **Positive MPL:** more labour raises output.
- **Diminishing MPL:** each additional unit of labour adds less output (holding K or T fixed).

Profit maximisation (competitive equilibrium):

$$w = P_M \cdot MPL_M(K, L_M) = P_F \cdot MPL_F(T, L_F)$$

Each sector hires labour until the value of its marginal product equals the wage.

These two conditions + $L_M + L_F = L$ determine L_M^* , L_F^* , and w^* .



The Specific Factors PPF: Concave, Not Linear

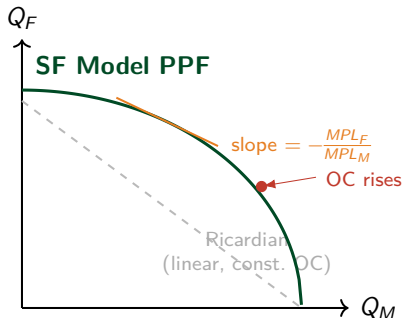
Why concave?

- When labour shifts from F to M, the first workers are highly productive in M (high MPL_M) but the last workers were still productive in F.
- As more labour moves to M, MPL_M falls and the lost food output per worker rises.
- Each extra unit of M comes at an *increasing* OC in terms of F.

Slope $= -MPL_F/MPL_M =$ opportunity cost of M.

As Q_M rises, more labour is in M, so MPL_M falls and MPL_F rises $\Rightarrow$ slope steepens.

Contrast with Ricardian: linear PPF (constant OC).



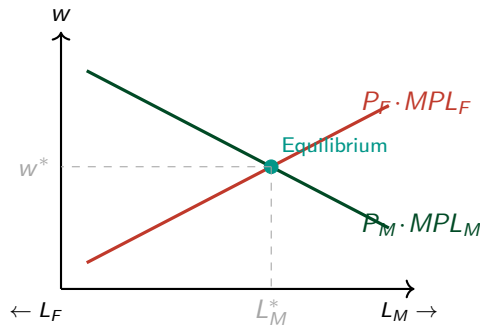
Labour Market Equilibrium: The VMPL Diagram

Equilibrium condition:

$$P_M \cdot MPL_M = P_F \cdot MPL_F = w$$

Reading the diagram:

- Horizontal axis: labour in M (left→right) and labour in F (right→left); total length = L .
- $P_M \cdot MPL_M$ slopes *down* as L_M rises (diminishing MPL in M).
- $P_F \cdot MPL_F$ slopes *down* as L_F rises — equivalently, slopes *up* as L_M rises (reading left to right).
- Crossing point: equilibrium (L_M^*, w^*) .



Worked Example: Specific Factors with Cobb-Douglas

Worked Example

Home produces Manufactures (M) and Food (F) with Cobb-Douglas production: $Q_M = K^{1/2}L_M^{1/2}$, $Q_F = T^{1/2}L_F^{1/2}$, $L_M + L_F = L$.

Endowments: $K = 144$, $T = 64$, $L = 100$. Prices: $P_M = 2$, $P_F = 1$.

Step 1 — Derive MPLs:

$$MPL_M = \frac{1}{2} K^{1/2} L_M^{-1/2} = \frac{6}{\sqrt{L_M}}$$

$$MPL_F = \frac{1}{2} T^{1/2} L_F^{-1/2} = \frac{4}{\sqrt{L_F}}$$

Step 2 — Equalize ($P_M \cdot MPL_M = P_F \cdot MPL_F$):

$$\frac{12}{\sqrt{L_M}} = \frac{4}{\sqrt{100 - L_M}}$$

Step 3 — Solve for L_M :

Square $\Rightarrow 144(100 - L_M) = 16 L_M$:

$$14400 = 160 L_M \Rightarrow \boxed{L_M = 90, L_F = 10}$$

Step 4 — Wage:

$$w = 2 \cdot \frac{6}{\sqrt{90}} = \frac{12}{\sqrt{90}} \approx 1.265$$

Worked Example: The VMPL Diagram

Home's VMPL curves ($K = 144$,
 $T = 64$, $L = 100$):

$$w_M(L_M) = P_M \cdot MPL_M = \frac{12}{\sqrt{L_M}}$$

$$w_F(L_M) = P_F \cdot MPL_F = \frac{4}{\sqrt{100 - L_M}}$$

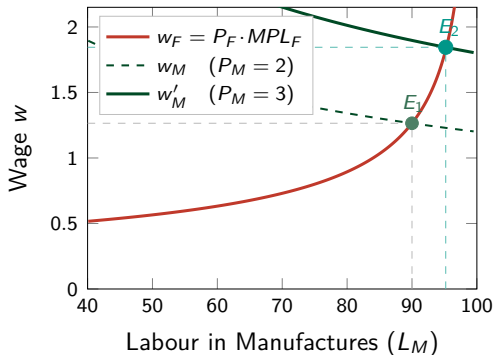
At $P_M = 2$: equilibrium $L_M = 90$,
 $w \approx 1.265$.

After P_M rises to 3:

$$w'_M = \frac{18}{\sqrt{L_M}}$$

New equilibrium: $L_M \approx 95.2$, $w \approx 1.845$.

Note: P_M rose by 50% ($2 \rightarrow 3$) but w rose



Effect of Trade: P_M Rises (Home Exports Manufactures)

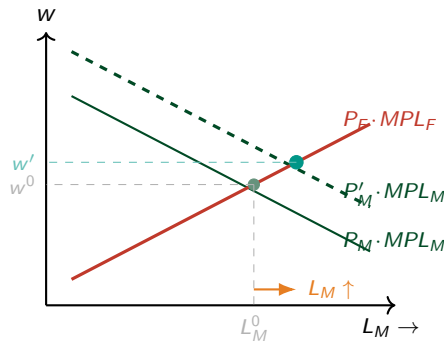
Suppose Home's CA is in manufactures.

Opening to trade raises P_M .

Sequence of effects:

1. $P_M \uparrow$ shifts $P_M \cdot MPL_M$ **up**.
2. At old wage, M now pays more $\Rightarrow$ workers move $M \leftarrow F$.
3. $L_M \uparrow$, $L_F \downarrow$.
4. Wage w rises — but **less than proportionally** to P_M (because MPL_M falls as more workers crowd into M).

Key result: $\hat{w} < \hat{P}_M$ (percentage change in wage is less than percentage change in P_M).



Why Labour's Real Wage Effect is Ambiguous

Nominal wage rose. Is labour better off?

After P_M rises: w rose, P_M rose, P_F unchanged.

Real wage in terms of M (what matters for workers who buy manufactures):

$$\frac{w}{P_M} \text{ fell}$$

Because $\hat{w} < \hat{P}_M$ — wage rose less than the price of manufactures.

Real wage in terms of F (what matters for workers who buy food):

$$\frac{w}{P_F} \text{ rose}$$

Because $\hat{w} > 0 = \hat{P}_F$ — wage rose while food price is unchanged.

The Ambiguity

Workers are **not unambiguously better off** or worse off — it depends on their **consumption basket**:

- Workers who mostly *buy goods* (urban manufacturing workers): real wage fell in M terms — **worse off**.
- Workers who mostly *eat* (agricultural workers): real wage rose in F terms — **better off**.

The General Rule

Labour is the *swing* factor: it gains in terms of the good whose price *did not* rise, and loses in terms of the good whose price rose.

Winners and Losers: Unambiguous Factor Returns

Capital owners (specific to M) — WIN

$$r_K = P_M \cdot MPK_M$$

$P_M \uparrow$ directly boosts revenue.

Extra labour flows into M $\Rightarrow MPK_M \uparrow$.

Double gain: real return to capital rises unambiguously.

Landowners (specific to F) — LOSE

$$r_T = P_F \cdot MPT_F$$

Labour left food sector $\Rightarrow L_F \downarrow \Rightarrow MPT_F \downarrow$.

P_F unchanged.

Double loss: real return to land falls unambiguously.

Labour — AMBIGUOUS

Wage w rises, but less than P_M rises.

- w/P_M falls (lost real purchasing power for manufactures).
- w/P_F rises (gained real purchasing power for food).

Net effect depends on the consumption basket.

Symmetric case ($P_F \uparrow$):

Land owners gain unambiguously.

Capital owners lose unambiguously.

Labour: ambiguous.

Summary: Winners and Losers

Factor	Sector	Effect of $P_M \uparrow$	Reason	Trade stance
Capital (K)	M (export)	Gain	Direct P_M gain + more labour in M	Favour free trade
Land (T)	F (import-comp.)	Lose	Labour left F; $MPT_F \downarrow$	Oppose free trade
Labour (L)	Both	Ambiguous	$w/P_M \downarrow$; $w/P_F \uparrow$	Divided

Political Economy Implication

Trade policy is contested because it creates **concentrated gains** for some and **concentrated losses** for others.

Capital owners in export sectors lobby *for* free trade; specific-factor owners in import-competing sectors lobby *against* it.

Why political opposition persists:

Gains are diffuse (many consumers, each gains a little). Losses are concentrated (few capital/land owners in import-competing sectors, each loses a lot).

Concentrated losers lobby harder.

The Magnification Effect (Jones, 1971)

Magnification Effect

If P_M rises by $x\%$ and P_F is unchanged:

$$\hat{r}_K > \hat{P}_M > \hat{w} > \hat{P}_F > \hat{r}_T$$

Factor returns change by *more* than goods prices.

Why? Capital gets a double gain:

1. *Direct*: P_M rises.
2. *Indirect*: More labour in M $\Rightarrow MPK_M \uparrow$.

Land suffers a double loss:

1. *No direct gain*: P_F unchanged.
2. *Indirect*: Labour left F $\Rightarrow MPT_F \downarrow$.

Alberta Oil Boom 2002–2014

When oil prices tripled, the SF model predicts:

Factor	Effect
Oil capital owners	Enormous gains
Alberta workers	Wage rise (+30–40%)
BC/ON manufacturers	Lost workers; costs $\uparrow$
Farmers (land)	Hurt (workers left)

Magnification: oil capital gains $>$ oil price gain; manufacturer capital loses $>$ it lost any direct revenue.

Does the Country Gain Overall?

The aggregate result:

- Trade raises **national income** at world prices — the economy-wide pie grows.
- But the gains are *unevenly distributed*: export-sector specific factors gain enormously; import-sector specific factors lose.
- **Kaldor-Hicks test**: the gainers' gains *exceed* the losers' losses, so **compensation is theoretically possible**.
- But compensation rarely occurs in practice. Trade adjustment assistance programs typically cover only a fraction of losses.

The Political Economy Lesson

Concentrated losses (50,000 steelworkers losing jobs) are politically visible and generate intense lobbying.

Diffuse gains (38 million consumers each saving \$200/car) are invisible.

This **asymmetry** explains persistent protection even when aggregate welfare rises.

CUSMA Dairy Tariffs

Canadian dairy supply management protects ~10,000 dairy farmers (specific capital in dairy) at the expense of 38 million Canadian consumers paying ~30% more for dairy products.

Comparative Advantage in the Specific Factors Model

Opportunity cost of M = slope of PPF:

$$OC_M = \left. \frac{dQ_F}{dQ_M} \right|_L = \frac{MPL_F}{MPL_M}$$

Reallocate dL : $dQ_M = MPL_M dL_M$,
 $dQ_F = -MPL_F dL_M$, giving
 $dQ_F/dQ_M = -MPL_F/MPL_M$.

With Cobb-Douglas $Q_M = K^{1/2} L_M^{1/2}$,
 $Q_F = T^{1/2} L_F^{1/2}$ (at equal labour split):

$$OC_M = \frac{MPL_F}{MPL_M} = \frac{T^{1/2}/\sqrt{L/2}}{K^{1/2}/\sqrt{L/2}} = \frac{T^{1/2}}{K^{1/2}}$$

Home ($K = 144$, $T = 64$):

$$OC_M = \sqrt{64}/\sqrt{144} = 8/12 = 2/3.$$

Foreign ($K^* = 64$, $T^* = 144$): $OC_M^* = 12/8 = 3/2$.

Factor Endowments Determine CA

Home is *capital-rich* ($K = 144 > 64$) and *land-poor* ($T = 64 < 144$).

Capital-richness lowers the OC of M — each unit of M forgoes less F.

⇒ Capital-rich countries export capital-intensive goods (manufactures).

⇒ Land-rich countries export land-intensive goods (food).

This is the **H-O theorem in embryo**: factor endowments determine comparative advantage.

Trade pattern: At world price $P_M/P_F \in (2/3, 3/2)$, Home exports M, Foreign exports F.

Canadian Application: Alberta Oil Boom

2002–2014: Global oil demand surged.

Canada's oil exports rose sharply.

Mapping to the SF model:

- **Good M:** oil & gas exports.
- **Good F:** manufactured goods (autos, steel).
- **Capital (K):** oil sands capital, drilling rigs (specific to energy sector).
- **Land (T):** land and natural resources in manufacturing regions (specific to autos/steel).
- **Labour (L):** mobile workers who moved to Fort McMurray.

SF Model Predictions Confirmed

Winners:

Energy capital owners (*huge* real gains via magnification). Alberta wages rose 35–40%.

Losers:

Ontario and BC manufacturers lost workers to Alberta (labour moved west). Manufacturer capital saw falling productivity as labour left.

Labour:

Ambiguous — Alberta workers gained; Ontario workers who stayed faced stagnant wages with rising goods prices.

This is Dutch Disease: oil boom hurts manufacturing capital.

Canadian Application: Ontario Manufacturing and Trade Shocks

Two import-competition episodes for Ontario:

1. China shock (2001–2015):

- Chinese manufactured goods entered Canadian market at lower prices.
- P_F (import-competing goods) fell — equivalent to adverse P_F shock.
- Ontario manufacturing capital (specific) lost *unambiguously*.
- Workers who stayed in manufacturing saw real wage gains (ambiguous channel — food prices also fell).

2. CUSMA auto content rules:

US tariff threat (2018–20) risked capital loss for auto-specific capital in Windsor-Oshawa-Oakville.

SF Model Lesson for Ontario

Ontario auto capital is *sector-specific*. Capital tied to auto assembly cannot quickly pivot to software or mining.

This explains why Ontario auto manufacturers (GM, Ford, Stellantis) intensely lobbied for CUSMA protections — their specific capital was at risk.

Policy implication: Even if overall Canada gains from CUSMA, sector-specific capital owners in exposed sectors lose, generating concentrated political opposition.

Canadian Application: Immigration as a Labour Supply Shock

Canada's immigration rate is among the highest in the OECD — $\sim 400,000$ permanent residents/year.

SF model analysis of a labour supply increase ($L \uparrow$):

- Both L_M and L_F absorb more workers.
- $MPL_M \downarrow$ and $MPL_F \downarrow \Rightarrow$ wages fall.
- But $MPK_M \uparrow$ (more labour with same K) $\Rightarrow r_K$ rises.
- And $MPT_F \uparrow \Rightarrow r_T$ rises.
- **Capital and land owners gain; native workers face wage pressure.**

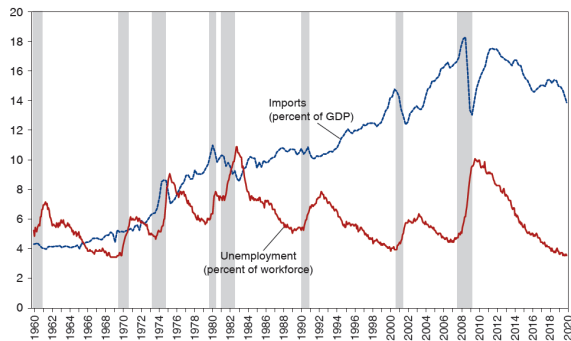
Canada's Immigration System

Canada targets **skill-complementary** immigration — many arrivals are in healthcare, STEM, and trades where labour is scarce.

Key distinction: High-skill immigrants *complement* native capital (raise MPK) more than they substitute for native labour. Low-skill immigrants compete more directly.

Net effect in Canada: capital-intensive sectors (finance, tech) benefit; wage effects in service/ construction sectors remain contested.

Evidence: US Unemployment and Import Penetration



What this figure shows:

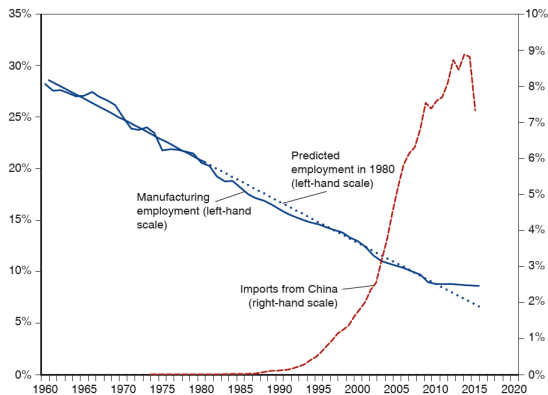
- Horizontal: import penetration ratio (imports/domestic consumption) by US industry.
- Vertical: unemployment rate in that industry.
- Industries with higher import penetration have **higher unemployment** — the SF model in action.

SF Model Prediction Confirmed

Workers in import-competing industries face job losses when trade opens. This is capital specificity at work: workers and capital in the wrong sector cannot instantly redeploy.

But *overall* unemployment need not rise — workers eventually find other jobs (mobile factor).

Evidence: The China Shock (Autor, Dorn & Hanson, 2013)



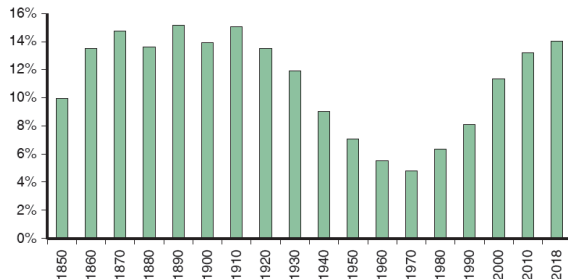
What this figure shows:

- US manufacturing employment (left axis) fell from $\approx 30\%$ to $\approx 15\%$ of the workforce, 1960–2020.
- Imports from China (right axis) rose from $\approx 1\%$ to $\approx 10\%$ over the same period.
- The inverse correlation is striking: as China's import share rose, US manufacturing employment fell.

SF Model Lesson

Losses in manufacturing were **persistent** (Autor, Dorn & Hanson, 2013): workers did not quickly re-employ at similar wages. This confirms labour is not perfectly mobile in the short run — the SF model's assumption is realistic.

Immigration Data: Canada Among Highest in the OECD



What this figure shows:

- Foreign-born share of the population across selected countries.
- **Switzerland, Australia, Canada:** very high (20–30%).
- **USA:** ~15% and rising.

Canada's Immigration Target

Canada admitted ~405,000 permanent residents in 2021 and over 400,000 each year thereafter — approximately 1% of population per year, among the highest rates in the world.

SF model: this labour supply increase *should* raise returns to Canadian capital and land while putting downward pressure on wages.

Short Run vs. Long Run: Bridge to Chapter 5

Feature	Short run (Ch. 4)	Long run (Ch. 5)
Capital	Specific to M	Mobile
Land	Specific to F	Mobile
Labour	Mobile	Mobile
Factors	3 (L, K, T)	2 (L, K)
PPF	Concave (smooth)	Concave (HO)
Winners	Clear (specific)	Via Stolper-Sam.

- Short run: a steel mill takes years to redeploy. A farm cannot quickly become a factory.
- Long run: capital depreciates and re-invests; land rezones; factor price equalisation emerges.
- Both time horizons matter: workers and capital owners care about the short run; economists focus on long-run efficiency.

Which Time Horizon Matters?

For policy analysis, both. Workers and capital owners in affected sectors care about the *short run* (their current income and assets). Policy-makers focused on long-run efficiency ignore short-run transitional losses.

The political conflict *arises from this mismatch*.

Timeline Analogy

After NAFTA (1994), Canadian auto parts manufacturers had perhaps 5–10 years to adjust as Mexican competition grew. This transition period determines whether the SF model or H-O model is more applicable.

Macro Connection: Sector-Specific Shocks and the Economy

The SF model connects to macroeconomics:

- **Commodity boom** (oil price rise): energy capital gains, manufacturing capital loses $\Rightarrow$ **Dutch Disease**. CAD appreciates, reducing manufactured export competitiveness.
- **Regional divergence**: oil boom concentrates activity in Alberta; manufacturing losses concentrated in Ontario/Quebec. Bank of Canada faces a difficult trade-off (one interest rate for very different regions).
- **Trade shock** $\rightarrow$ **GDP composition**: sector-specific effects change the mix of consumption, investment, and exports — affecting aggregate demand and the current account.

SF Model and Bank of Canada

2002–2014: Alberta oil boom raised wages and incomes there but hurt Ontario manufacturing.

The Bank of Canada set *one* interest rate for the whole country. A rate cut to help Ontario would overheat Alberta; a rate hike to cool Alberta would hurt Ontario.

This **regional asymmetry of trade shocks** is a recurring challenge for Canadian monetary policy — rooted in the Specific Factors model.

Key Takeaways

1. The SF model has **three factors**: labour (mobile), capital (specific to M), land (specific to F).
2. Equilibrium: $P_M \cdot MPL_M = P_F \cdot MPL_F = w$.
3. A rise in P_M **unambiguously benefits** capital (specific to M) and **unambiguously harms** land (specific to F).
4. **Labour is ambiguous**: real wage rises in terms of F, falls in terms of M.
5. **Magnification effect**: factor returns change *more* than goods prices.
6. Countries gain in aggregate from trade, but distribution is unequal — political conflict is the predictable result.
7. Canada: Alberta oil capital won, Ontario manufacturer capital lost; immigration raises capital returns.
8. **Short run** → **Ch. 4**; **Long run** → **Ch. 5**.

Next Lecture: Chapter 5

The Heckscher-Ohlin Model

Long run: both capital and labour are mobile between sectors.

Factor endowments (not specific factors) determine comparative advantage.

Factor price equalisation, Stolper-Samuelson theorem, Rybczynski theorem, Leontief Paradox.

Discussion Questions

Discussion

1. In the Canadian forestry sector, who wins and who loses when global lumber prices rise? Map each party to the specific factors (capital, land, labour).
2. The US imposed tariffs on Canadian softwood lumber. Using the SF model, show *who in the US gains* (capital in US forestry) and *who in Canada loses* (Canadian forestry capital).
3. If automation destroys manufacturing jobs, is this similar to import competition? How would the SF model describe the distributional effects of automation?
4. Should Canada compensate the losers from trade? What policy instruments could achieve this?

Key readings:

- Jones, R.W. (1971), "A Three-Factor Model," *Trade, Balance of Payments and Growth*.
- Autor, D., Dorn, D., & Hanson, G. (2013), "The China Syndrome," *American Economic Review*.
- Borjas, G.J. (1995), "The Economic Benefits from Immigration," *Journal of Economic Perspectives*.

Next Lecture

Ch. 5 — Heckscher-Ohlin: capital and labour, long-run factor mobility, four major theorems.